

Tutorial set 2

1. Consider the wave function

$$\psi(x, t) = Ae^{-\lambda|x|}e^{-i\omega t}$$

where A , λ and ω are positive real constants. (a) Normalize ψ . (b) Determine the expectation values of x and x^2 .

2. A particle is represented (at time $t = 0$) by the wave function

$$\psi(x, 0) = A(a^2 - x^2), \quad \text{if } -a \leq x \leq a$$

and zero elsewhere. (a) Determine the normalization constant A . (b) What is the expectation value of x at time $t = 0$? (c) What is the expectation value of p at time $t = 0$? (d) Find the expectation values of x^2 and p^2 . (e) Find the uncertainty in x and p . (f) Check that the results are consistent with uncertainty principle.

3. A particle in infinite square well has its initial wave function an even mixture of two stationary states

$$\psi(x, 0) = A[\psi_1(x) + \psi_2(x)]$$

(a) Normalize $\psi(x)$. (b) Find $\psi(x, t)$ and $|\psi(x, t)|^2$. (c) Compute the expectation value of x . Note that it oscillates in time. What is the angular frequency of oscillation? What is the amplitude of oscillation? (d) Compute the expectation value of p . (e) If you measure the energy of the particle, what values might you get? What is the probability of getting each of them? Find the expectation value of H . How does it compare with E_1 and E_2 ?

4. Consider a particle of mass m and energy $E > 0$ held in a potential $-V_0\delta(x - a)$. (a) Integrate the Schrodinger equation between $a - \epsilon$ and $a + \epsilon$. Taking the limit $\epsilon \rightarrow 0$, show that the derivative of the eigenfunction has a discontinuity at $x = a$ and determine it. (b) Obtain the eigenfunction in both regions, $x > a$ and $x < a$.

5. A particle in the infinite square well has the wave function

$$\psi(x, 0) = A \sin^3\left(\frac{\pi x}{a}\right) \quad 0 \leq x \leq a$$

Determine A , find $\psi(x, t)$ and calculate the expectation value of x , as a function of time. What is the expectation value of energy?

6. Imagine a bead of mass m that slides frictionlessly around a circular wire of circumference L (this is just a free particle, except $\psi(x+L) = \psi(x)$). Find the stationary states and the corresponding allowed energies.